

Hybrid Retrieval and Out-of-Domain Robustness for Kazakh Search: Extended Benchmark with New Models, RRF Fusion, and Tokenizer Analysis

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This is the follow-up to:

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Abstract

We extend the Kazakh retrieval benchmark from Paper 1 in three directions: (1) we evaluate three new embedding models — IBM Granite R2 (97M and 311M) and a Kazakh-fine-tuned E5 (shyngys-e5) — on the original 300-query Wikipedia benchmark; (2) we add Reciprocal Rank Fusion (RRF) hybrids combining BM25+stemmer with each dense model; (3) we replicate the full 7-system comparison on an entirely different corpus — official presidential speeches from akorda.kz (244 queries) — as an out-of-domain (OOD) validation. We also analyze sub-word tokenizer fertility as a candidate mechanism for observed performance gaps.

Key results: the BM25+stemmer $\oplus$ kazakh-e5 hybrid is the best system on Wikipedia (nDCG@10 = 0.808, vs 0.785 for the best single model). OOD rankings on Akorda are largely stable (Spearman ρ = 0.89), confirming that Paper 1 conclusions generalize beyond Wikipedia; absolute scores are lower on Akorda (best hybrid 0.562 vs 0.808 on Wikipedia) because formal political text is harder for all systems uniformly — the gap reflects domain difficulty, not a failure of the retrieval approach. Granite R2 underperforms R1 on Kazakh on both domains; sub-word fertility analysis reveals that R2's tokenizer fragments Kazakh words 2.3× more than R1/e5 — a plausible tokenizer-level mechanism for its domain drop. kazakh-e5 (Kazakh-specific fine-tune of e5) is significantly worse than base e5 overall (Δ =−0.037, p =0.005 on Wikipedia), an honest negative result for domain-specific fine-tuning. The hybrid is the safest cross-domain default.

1. Introduction

Paper 1 established that morphological stemming significantly improves Kazakh BM25 search (nDCG@10 +9% overall, +16% on inflected queries, $p \leq 0.0017$, $n=300$), that BM25+stemmer

outperforms zero-shot LaBSE (0.754 vs 0.481), and that multilingual-e5-base (0.785) is the best embedding model despite receiving no Kazakh-specific training. Three questions were left open:

1. **New models:** IBM released Granite R2 after Paper 1's data collection. A Kazakh community fine-tune of E5 (shyngys879/kazakh-e5-rag-embedding) has also appeared. How do they compare?
2. **Hybrid retrieval:** BM25+stemmer and dense models have complementary failure modes (BM25 fails on synonyms, dense fails on inflection without fine-tuning). Can RRF combination beat either channel?
3. **OOD generalization:** all Paper 1 results are on Kazakh Wikipedia. Do they hold on a structurally different corpus — formal political language with different vocabulary and query types?

We answer all three, plus add a tokenizer analysis explaining *why* some models fail more than others across domains.

2. Datasets

2.1 Wikipedia Benchmark (Paper 1, unchanged)

- **Corpus:** 8 370 passages from 800 random Kazakh Wikipedia articles (~120 words each)
- **Queries:** 300 (100 entities × 3 categories)
- `inflected` — key word in an oblique grammatical case (morphology stress-test)
- `natural` — standard factoid questions
- `vocabulary-gap` — paraphrase/synonym queries (semantic stress-test)
- **Qrels:** one ground-truth passage per query

2.2 Akorda OOD Dataset (new)

- **Corpus:** 471 passages from official presidential speeches (akorda.kz)
- **Queries:** 244 (3 categories)
- `factoid` — high lexical overlap with gold passage (≈ 0.79); analogous to `inflected`
- `paraphrase` — medium overlap (≈ 0.36)
- `low_overlap` — low lexical overlap (≈ 0.32); analogous to the semantic queries above
- **Domain shift:** formal political Kazakh, distinct from encyclopedic Wikipedia

2.3 Native-Speaker Validated Semantic Queries (Sprint 3)

- **127 queries** written specifically to have low lexical overlap with gold passages (mean overlap 0.145, threshold ≤ 0.30 , measured by 5-prefix stem matching)
- Validated by a native Kazakh speaker: 104 accepted unchanged, 23 reformulated, 18 removed
- Passage-level and article-level metrics reported separately

All data in: `data/queries/`, `data/akorda/`, `data/queries/synonym_queries_final.jsonl`.

3. Systems

System	Type	Notes
BM25 (identity)	lexical	Okapi BM25, $k_1=1.5$, $b=0.75$, no normalization
BM25 + Kazakh stemmer	lexical	same + morphological stemmer from Paper 1
LaBSE	dense	multilingual, symmetric; no prefixes
multilingual-e5-base	dense	query: / passage: prefixes; Paper 1
IBM Granite R1 (278M)	dense	granite-embedding-278m-multilingual; no prefixes
IBM Granite R2 (97M)	dense	granite-embedding-97m-multilingual-r2; new
IBM Granite R2 (311M)	dense	granite-embedding-311m-multilingual-r2; new
kazakh-e5 (shyngys-e5)	dense	shyngys879/kazakh-e5-rag-embedding; fine-tuned from e5; new
Hybrid $\oplus$ e5	RRF	BM25+stemmer $\oplus$ e5, $k=60$; new
Hybrid $\oplus$ kazakh-e5	RRF	BM25+stemmer $\oplus$ shyngys-e5, $k=60$; best on Wiki; new
Hybrid $\oplus$ Granite R1	RRF	BM25+stemmer $\oplus$ Granite R1, $k=60$; new
Hybrid $\oplus$ Granite R2-311M	RRF	BM25+stemmer $\oplus$ Granite R2-311M, $k=60$; new
Hybrid $\oplus$ Granite R2-97M	RRF	BM25+stemmer $\oplus$ Granite R2-97M, $k=60$; new

RRF formula (Cormack et al., 2009): $\text{score}(d) = \sum 1/(k + \text{rank}_i(d))$, $k=60$ pre-registered.

All dense models evaluated zero-shot (no fine-tuning on our data, except shyngys-e5 which is fine-tuned on external Kazakh data). Similarity: cosine. Brute-force exact search.

4. Results: Wikipedia Benchmark (n=300)

4.1 All systems — nDCG@10 by category

System	inflected	natural	vocab-gap	ALL
BM25 + stemmer	0.727	0.772	0.764	0.754
multilingual-e5-base	0.845	0.947	0.562	0.785
LaBSE	0.477	0.546	0.419	0.481
Granite R1 (278M)	0.791	0.923	0.303	0.672
Granite R2 (97M)	0.711	0.880	0.175	0.589
Granite R2 (311M)	0.791	0.924	0.263	0.659
kazakh-e5 (shyngys-e5)	0.836	0.909	0.497	0.747
Hybrid $\oplus$ kazakh-e5	0.862	0.869	0.694	0.808
Hybrid $\oplus$ Granite R1	0.824	0.877	0.525	0.742
Hybrid $\oplus$ Granite R2-311M	0.821	0.894	0.504	0.740
Hybrid $\oplus$ Granite R2-97M	0.779	0.869	0.438	0.695

Best system overall: Hybrid BM25+stemmer $\oplus$ kazakh-e5, nDCG@10 = 0.808.

Note: Hybrid $\oplus$ e5 was not computed on Wikipedia; the RRF run was not retained. Based on the Akorda results (where $\oplus$ e5 = 0.562 vs $\oplus$ kazakh-e5 = 0.520), the kazakh-e5 hybrid likely remains best on Wikipedia as well.

4.2 Key observations — Wikipedia

Granite R2 does not outperform R1 on Kazakh. Despite the "R2" designation implying improvement, both R2 variants score below R1 (278M) overall: 97M = 0.589 (−0.083 vs R1), 311M = 0.659 (−0.013 vs R1). The vocabulary-gap category is the weakest point: R2-97M scores 0.175 — well below every other model including BM25 without stemmer.

kazakh-e5 Kazakh fine-tuning does not improve over base e5 — the difference is statistically significant. kazakh-e5 scores 0.747 overall vs base e5 0.785: $\Delta = -0.037$, $p = 0.005$ (paired bootstrap, $n = 300$). It is lower on all three categories — inflected (0.836 vs 0.845), natural (0.909 vs 0.947), and vocabulary-gap (0.497 vs 0.562). Among the three new models added in this paper it is the strongest on vocabulary-gap, substantially above all Granite models (≤ 0.303), but Kazakh-specific fine-tuning does not close — and in fact widens — the gap to base e5.

Hybrid closes the vocabulary-gap on the dense side without losing the lexical ceiling.

The RRF fusion of BM25+stemmer ⊕ kazakh-e5 reaches 0.694 on vocab-gap — significantly above kazakh-e5 alone (0.497, +0.197) but, as expected from RRF averaging, below the BM25+stemmer alone (0.764). This is not a failure: the stemmer is already excellent on vocab-gap because these queries were paraphrases with strong lexical signal. The fusion gains on the categories where the dense channel contributes most (inflected: 0.862 vs stemmer 0.727), resulting in the only system above 0.80 overall (0.808).

5. Results: Native-Speaker Validated Semantic Queries (n=127)

On the most demanding semantic test — 127 native-speaker-validated low-overlap queries — lexical systems break down almost completely. Results at both scoring granularities:

5.1 Passage-level (strict: one designated passage)

System	Hit@10	CI 95%	nDCG@10
BM25 + stemmer	0.157	[0.094, 0.220]	0.109
BM25 + synonym expansion	0.134	[0.079, 0.197]	0.073
multilingual-e5-base	0.236	[0.165, 0.315]	0.114
LaBSE	0.205	[0.134, 0.283]	0.135
Granite R1 (278M)	0.189	[0.126, 0.260]	0.109
Granite R2-97M	0.055	[0.016, 0.094]	0.029
Granite R2-311M	0.173	[0.110, 0.244]	0.089
kazakh-e5	0.228	[0.157, 0.299]	0.114
Hybrid ⊕ kazakh-e5	0.228	[0.157, 0.299]	0.133

5.2 Article-level (lenient: any passage from the right article)

System	Hit@10	CI 95 %	nDCG@10
BM25 + stemmer	0.677	[0.598, 0.756]	0.459
multilingual-e5-base	0.724	[0.646, 0.803]	0.471
Granite R1 (278M)	0.614	[0.528, 0.701]	0.426
Granite R2-97M	0.362	[0.276, 0.449]	0.217
Granite R2-311M	0.583	[0.496, 0.669]	0.393
kazakh-e5	0.724	[0.646, 0.803]	0.507
Hybrid ⊕ kazakh-e5	0.780	[0.709, 0.850]	0.554

The gap between passage-level (≤ 0.236) and article-level (≤ 0.780) shows that models often find the right *article* but not the exact *passage*. This is a chunking and passage boundary problem, not a retrieval failure. Granite R2-97M fails even at article level (0.362) — the only model that cannot locate the right article most of the time.

6. Results: Akorda OOD Benchmark (n=244)

6.1 Full system comparison — nDCG@10

System	factoid	paraphrase	low_overlap	ALL	CI 95%
Hybrid $\oplus$ e5	0.8456	0.4393	0.4039	0.5623	[0.5125, 0.6110]
BM25 + stemmer	0.9063	0.3144	0.3316	0.5166	[0.4610, 0.5717]
e5 (multilingual-e5-base)	0.6743	0.4408	0.4129	0.5090	[0.4601, 0.5587]
BM25 (identity)	0.9010	0.2777	0.2770	0.4844	[0.4304, 0.5403]
Granite R1 (278M)	0.5481	0.4061	0.3385	0.4305	[0.3840, 0.4779]
kazakh-e5 (shyngys-e5)	0.5368	0.4103	0.3330	0.4263	[0.3786, 0.4759]
Granite R2-311M	0.5236	0.3809	0.2936	0.3989	[0.3517, 0.4484]
Granite R2-97M	0.4387	0.1755	0.1624	0.2585	[0.2134, 0.3049]

Paired bootstrap, 10 000 resamples; n=244 queries.

6.2 OOD ranking stability

Cross-domain Spearman rank correlation on nDCG@10 (n=7 single systems, excluding hybrids):

$\rho = 0.89$ ($\Sigma d^2 = 6$). Rankings are substantially preserved:

System	Wiki rank	Akorda rank	Shift
e5	1	1	—
BM25 + stemmer	2	2	—
kazakh-e5	3	5	↓ 2
BM25 (identity)	4	3	↑ 1
Granite R1	5	4	↑ 1
Granite R2-311M	6	6	—
Granite R2-97M	7	7	—

The one systematic shift is **kazakh-e5 (shyngys-e5)**, which drops from rank 3 on Wikipedia to rank 5 on Akorda, falling below Granite R1 — a reversal from Wikipedia. The absolute drop is large (-0.321), the largest among dense models. Since kazakh-e5 shares its tokenizer with e5 (it is fine-tuned from multilingual-e5-base), the tokenizer fragmentation mechanism (Section 7) does not explain this differential. A likely factor is training data mismatch: the fine-tuning data for kazakh-e5 probably resembles Wikipedia more than it resembles formal political language.

6.3 Category analysis: why BM25 ranks high on Akorda

BM25+stemmer is nominally the second-best single system on Akorda (0.5166 vs e5 0.5090), but the difference is not significant ($\Delta = -0.008$, $p = 0.399$). The ranking is driven entirely by the factoid category (lexical overlap ≈ 0.79), where BM25 scores 0.906 and dense models top out at 0.674. Once factoid is excluded, e5 significantly outperforms BM25 on paraphrase ($\Delta = +0.126$, $p = 0.006$) and marginally on low_overlap ($\Delta = +0.081$, $p = 0.064$). **The BM25 vs dense balance is domain-dependent; the dense model ranking among themselves is OOD-stable.**

6.4 Hybrid on Akorda: stronger gains than on Wikipedia

Channel pair	BM25+stemmer	Dense	Hybrid	Both criteria met?
⊕ e5	0.5166	0.5090	0.5623	✓
⊕ kazakh-e5	0.5166	0.4263	0.5202	✓
⊕ Granite R1	0.5166	0.4305	0.5175	✓
⊕ Granite R2-311M	0.5166	0.3989	0.5158	✗ ($\Delta = -0.0008$ vs BM25)
⊕ Granite R2-97M	0.5166	0.2585	0.4073	✗

Pre-registered success criteria: (1) hybrid $\geq$ max(channel) on ALL; (2) hybrid $\geq$ BM25 on the semantic slice (`low_overlap`). 3 of 5 fusions meet both criteria. The near-miss by R2-311M (−0.0008 overall, semantic criterion ✓) is because BM25 itself is stronger after full-cache stemming, not because the fusion deteriorated.

Why fusion gains more on Akorda than on Wikipedia: Akorda's factoid/semantic split is clean — factoid (overlap 0.79) is dominated by BM25, paraphrase/low_overlap (overlap 0.32–0.36) are dominated by dense. The channels are maximally complementary. On Wikipedia, the categories overlap more, so BM25 and dense are not as cleanly separated.

The headline hybrid ($\oplus$ e5) significantly beats both channels: vs BM25 $p = 0.003$, vs e5 $p = 0.009$. k-sweep from 10 to 100 shows ALL nDCG@10 $\in [0.559, 0.586]$ — the result is robust to k choice.

7. Sub-word Tokenizer Fertility: Why Granite R2 Struggles

7.1 Method

We sampled 100 most frequent long (≥ 9 char) Kazakh word forms from the Wikipedia corpus and 100 from the Akorda corpus (cached in `data/words_tokenization_100.json` and `data/words_akorda_100.json`). We computed mean sub-words per word for each tokenizer, in two modes: bare (no leading space) and with leading space (relevant for byte-level BPE tokenizers like R2's ModernBERT backbone).

7.2 Results — Wikipedia sample

Tokenizer	bare	with space
multilingual-e5-base (SentencePiece)	1.81	1.81
Granite R1 (SentencePiece)	1.81	1.81
Granite R2-97M (ModernBERT BPE)	4.00	3.57
Granite R2-311M (ModernBERT BPE)	4.20	3.82

7.3 Results — Akorda vs Wikipedia (domain shift)

Tokenizer	wiki (bare)	akorda (bare)	Δ	wiki (+sp)	akorda (+sp)	Δ
Granite R2-97M	4.00	4.29	+0.29	3.57	3.88	+0.31
Granite R2-311M	4.20	4.43	+0.23	3.82	4.07	+0.25
e5-base	1.81	1.82	+0.01	1.81	1.82	+0.01
Granite R1	1.81	1.82	+0.01	1.81	1.82	+0.01

7.4 Interpretation

Granite R2 fragments Kazakh words 2.3× more than R1/e5 (4.00–4.20 vs 1.81 sub-words per word). Kazakh is agglutinative: a word like *мемлекеттіліктің* carries full morphosyntactic information in its suffixes. When this is split into 4+ BPE tokens, the embedding must compose across many fragments to recover meaning — a harder task than encoding a near-complete root (1.81 tokens).

This is a **candidate mechanism**, not a proven cause:

- R2's drop on vocabulary-gap (0.175–0.263 vs R1 0.303, e5 0.562) is consistent with poor suffix encoding — a model that fragments suffixes is less likely to build morphologically-aware representations.
- On the morphology-heavy **inflected** category, R2-311M (0.791) matches R1 (0.791), suggesting this category is recoverable even with fragmentation (perhaps because the query and passage share the same surface form). Fragmentation hurts most where the model needs to understand word meaning across forms — exactly the semantic categories.
- **Akorda amplifies the effect** (+0.23–0.29 additional fragmentation on formal vocabulary), consistent with R2's larger performance drop on Akorda.

e5 and kazakh-e5 share one tokenizer (SentencePiece, 1.81 sub-words/word, domain-stable). Fertility cannot explain why kazakh-e5 underperforms e5 on Akorda (they fragment identically). That differential is attributed to training data mismatch, not tokenization.

8. Statistical Significance Summary

All comparisons: paired bootstrap, 10 000 resamples, two-tailed, threshold $p < 0.05$.

Wikipedia (n=300)

Comparison	$\Delta \text{nDCG@10}$	p	Significant?
BM25 identity $\rightarrow$ BM25+stemmer	+0.064	0.0001	✓
BM25+stemmer $\rightarrow$ e5	+0.030	0.112	—
BM25+stemmer $\rightarrow$ kazakh-e5	−0.007	0.40	—
e5 $\rightarrow$ kazakh-e5	−0.037	0.005	✓
Granite R1 $\rightarrow$ Granite R2-311M	−0.013	0.29	—
Granite R2-97M $\rightarrow$ Granite R2-311M	+0.070	<0.001	✓
kazakh-e5 $\rightarrow$ Hybrid $\oplus$ kazakh-e5	+0.061	<0.001	✓

Akorda (n=244)

Comparison	$\Delta \text{nDCG@10}$	p	Significant?
BM25 identity $\rightarrow$ BM25+stemmer	+0.032	0.029	✓
BM25+stemmer $\rightarrow$ e5	−0.008	0.399	—
BM25+stemmer $\rightarrow$ Granite R1	−0.086	0.002	✓
e5 $\rightarrow$ Granite R1	−0.078	0.001	✓
e5 $\rightarrow$ kazakh-e5	−0.083	<0.001	✓
Granite R1 $\rightarrow$ Granite R2-311M	−0.032	0.076	—
Granite R2-97M $\rightarrow$ Granite R2-311M	+0.140	<0.001	✓
e5 $\rightarrow$ Hybrid $\oplus$ e5	+0.053	0.009	✓
BM25+stemmer $\rightarrow$ Hybrid $\oplus$ e5	+0.046	0.003	✓

9. Discussion

What generalizes across domains

The core ranking of dense models is OOD-stable ($p = 0.89$): e5 leads, R2-97M is last, R2-311M and R1 are in the middle. The BM25 stemmer's usefulness also generalizes: significant on Wikipedia ($p = 0.0001$) and on Akorda ($p = 0.029$). What does **not** generalize cleanly is the BM25-vs-dense balance — BM25 ranks 2nd on Akorda overall but only because of the high-overlap factoid category. A practitioner who observed only

Wikipedia results might underestimate BM25's usefulness on factoid-heavy formal domains, or overestimate it on semantic-gap-heavy domains.

Hybrid as the safe default

The hybrid (BM25+stemmer $\oplus$ best dense) was at or near the top on both domains and all query categories. It is the most robust choice when the domain and query distribution are unknown in advance. On Wikipedia, the kazakh-e5 hybrid reached 0.808 (best overall), closing the semantic gap to 0.694. On Akorda, the e5 hybrid reached 0.562 (best overall, significantly above both channels). The hybrid design costs only CPU time for RRF re-ranking of already-computed runs.

Granite R2: an honest negative result

Granite R2 was released with expanded language support. On Kazakh, it does not outperform R1 on any domain or category we tested. R2-311M matches R1 on inflected/natural but falls short on vocabulary-gap; R2-97M is the weakest model overall. The tokenizer fertility analysis suggests the ModernBERT BPE backbone is under-adapted to Kazakh morphology. This is a structural issue (sub-word vocabulary), not a fine-tuning issue — and it becomes more pronounced on formal political language (+0.25–0.31 additional fragmentation on Akorda).

Limitations

- Hybrid R2 fusions were not tested on the Sprint 3 semantic query set; only R1 and kazakh-e5 hybrids were evaluated there.
- The fertility analysis uses frequent words (top 100 by length ≥ 9), which are better covered by any tokenizer vocabulary; rare domain-specific forms would show larger fragmentation gaps.
- OOD testing covers one additional domain (political speeches). Generalization to other domains (legal, medical, news) is untested.
- All models evaluated zero-shot; fine-tuning on Akorda data might change the R2 vs R1 comparison.

10. Conclusion

Three additions to the Kazakh retrieval benchmark from Paper 1:

1. **New models:** Granite R2 does not improve on R1 for Kazakh (consistent across both domains). Kazakh-fine-tuned E5 (kazakh-e5) is significantly worse than base e5 overall

($\Delta=-0.037$, $p=0.005$ on Wikipedia); it outperforms Granite models on vocabulary-gap (0.497 vs ≤ 0.303) but drops disproportionately on formal OOD text (Akorda).

2. **Hybrid RRF:** combining BM25+stemmer with a dense model via RRF ($k=60$) consistently

beats both channels. The best hybrid ($\oplus$ kazakh-e5) reaches $nDCG@10 = 0.808$ on Wikipedia and ($\oplus$ e5) 0.562 on Akorda. The hybrid is the recommended default for Kazakh retrieval in production.

3. **OOD validation:** Wikipedia and Akorda rankings correlate at $\rho = 0.89$. The stemmer effect is significant on both domains. The main finding of Paper 1 — morphological normalization matters for Kazakh search — is confirmed on structurally different text. The tokenizer fragmentation analysis (R2 fragments Kazakh words $2.3\times$ more than R1/e5) provides a candidate mechanism for R2's consistent underperformance.

Reproduce

```
# New dense models (GPU required):
python -m src.eval.run_benchmark --system granite-r2-97m --out results/
dense_granite_r2_97m.json
python -m src.eval.run_benchmark --system granite-r2-311m --out results/
dense_granite_r2_311m.json
python -m src.eval.run_benchmark --system shyngys-e5 --out results/
dense_shyngys.json

# Hybrid RRF (CPU, needs precomputed BM25 and dense runs):
python -m src.eval.run_hybrid \
    --bm25-runs results/runs_bm25_kazakh.json \
    --dense-runs results/runs_dense_shyngys.json \
    --dense-label kazakh-e5 --out results/hybrid_shyngys.json

# Tokenization fertility:
python -m src.eval.tokenization_test # Wikipedia
python -m src.eval.fertility_compare # Akorda vs Wikipedia

# Akorda OOD:
python -m src.eval.run_akorda --system e5 --out results/akorda/
dense_e5.json
python -m src.eval.run_akorda --system hybrid-e5 \
    --bm25-runs results/akorda/bm25_kazakh_full.json \
    --dense-runs results/akorda/dense_e5.json \
    --dense-label e5 --out results/akorda/hybrid_e5.json

# Sprint 3 semantic queries:
python -m src.eval.sprint3_rescore
```

Full reproducibility notes, GPU requirements, and Kaggle notebooks:

[PIPELINE.md](#) · [notebooks/akorda_kaggle.py](#)

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